

The Impact of Weather Shocks on Violent and Property Crimes in Jamaica

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Abstract

Developing countries face the largest exposure to the negative effects of climate change. However, as temperature and rainfall patterns change, we have a limited understanding of their impact on these countries and the mitigation strategies that may be needed. In this paper, we utilize administrative panel data to examine the impact of weather shocks on violent and property crimes in Jamaica. We find strong evidence that a one-standard-deviation increase in the daily temperature (2°C) increases violent crime by 3.67 percent, due to an increase in the number of murders (3.44 percent), shootings (7.53 percent), and cases of aggravated assault (6 percent). However, our results suggest that temperature changes have no statistical impact on property crime. In addition, we find that a one-standard-deviation increase in rainfall (2 mm) reduces crimes such as shootings (1.53 percent), break-ins (2.27 percent), and larcenies (3.85 percent), but it has a minimal impact on other categories of crime.

JEL classification: K32, K42, Q54, Q58, I18

Keywords: crime, temperature, rainfall, climate

1. Introduction

Rising global temperatures, combined with widespread changes in weather patterns, have resulted in more extreme weather events such as heat waves, severe storms, extended droughts, and hurricanes over the last few decades (Ritchie, Roser, and Rosado 2020). These effects are likely to become more frequent and intense as the countries that are least vulnerable to the effects of climate change account for the highest greenhouse gas emissions.¹ This has resulted in lower-income developing countries bearing the

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- 1 For instance, 20 of the 36 countries with the highest emissions are among the least vulnerable to the negative effects of climate change. On the other hand, the countries that are most vulnerable to climate change have the least responsibility for global emissions and they suffer low economic growth and other severe negative climate change impacts (Althor, Watson, and Fuller 2016).

brunt of the adverse consequences of rising temperature, though they possess lower emission levels.² For instance, higher temperatures have been shown to damage health, lower agricultural productivity, and reduce per capita output in countries such as Bangladesh, Haiti, and Gabon (Lynch, Stretesky, and Long 2020; Acevedo et al. 2020). However, despite the documented evidence that climate change has caused more volatile and extreme weather conditions, not much is known about how this will affect individuals' propensity towards criminal activities in developing countries.

This paper examines the impact of weather shocks on incidents of violent and property crimes. The analysis uses a novel police-division-by-day-level panel that is constructed using administrative crime and weather data from multiple public agencies in Jamaica. This country is an interesting case to study because it is consistently ranked among the top 10 most violent countries in the world and because it is quite vulnerable to the negative effects of climate change with a risk assessment score which ranks it 47th out of 191 countries (Mertz et al. 2009; Rahim et al. 2023). This exposure partly exists because Jamaica and other developing countries currently lack the infrastructure that can mitigate the effects of extreme weather changes, and they face a significant challenge in financing longer-term climate change adaptation strategies.³ Our empirical approach examines how changes in the weather impact crime by leveraging the random variation in temperature and rainfall (shocks) that occurs within a geographical area, conditioning on various socio-economic covariates, day-of-the-year, day-of-the-week, month, and year fixed effects. We also show that our results are not sensitive to various model specifications and robustness checks.

This study is one of the few to examine the impact of weather shocks on individuals' behavior and propensity towards various crimes in a developing country.⁴ For instance, Baysan et al. (2019) and Garg, McCord, and Montfort (2020) examine the effect of temperature on violence in Mexico and found that higher temperature is associated with an increase in drug-related killings and homicides. Similarly, Blakeslee et al. (2021) use a unique police-station-by-day-level panel to examine the impact of weather on violent and property crimes in the Indian state of Karnataka. This study is noteworthy as it is the only one that has examined the relationship between temperature, rainfall, and crime in a developing country using high-frequency data at a disaggregated level, while focusing on a wide array of violent and property crimes.

- 2 Studies have shown that a 4 percent increase in temperature in developing countries results in a reduction of per capita output by up to 1.5 percent—a loss that may persist for at least 7 years (Burke, Hsiang, and Miguel 2015; Acevedo et al. 2020).
- 3 As temperature increases, a growing number of individuals will need access to air conditioning technology to avoid heat-related stress. Pavanello et al. (2021) argued that air-conditioning demand could increase between 2-fold and 16-fold within the next two decades. They found that the air-conditioning (AC) ownership rate is quite low in several developing countries such as India (12 percent), Mexico (14 percent), Indonesia (8 percent), and Brazil (20 percent). Similarly, AC ownership in Jamaica was last estimated at 3 percent in 2006, with average commercial and residential demand estimated at about 50,000 each year between 2013 and 2018 (Binger 2011; Japan Refrigeration and Air Conditioning Industry Association 2019). In comparison, AC ownership in the United States was 77 percent in 2001 and it grew to 88 percent in 2020 (Ross and McNary 2023). This evidence suggests that, unlike developed countries, developing countries currently lack the infrastructure to mitigate the effects of severe weather events.
- 4 Focusing on the impact of weather changes in developing countries is important because they have fewer adaptation technologies and most are located in the tropical and equatorial regions, which are prone to hotter average temperatures (Sachs 2001; Blakeslee et al. 2021). Due to these differences, the estimates that are reported in the literature for developed countries may not be representative of the reality of developing nations. See fig. S1.1 in the supplementary online appendix for a description of the average temperature of countries in the tropical region relative to the rest of the world. Other tropical countries in the Americas with similar temperatures to Jamaica include Brazil, Bolivia, Columbia, Costa Rica, Dominican Republic, Ecuador, Guatemala, Honduras, Panama, Peru, Nicaragua, and Venezuela (Racke et al. 1997).

Similar to [Blakeslee et al. \(2021\)](#), we utilize a novel incident-level administrative data set to examine the impact of temperature and rainfall shocks on crime in Jamaica. These data are used to create a unique police-division-by-day-level panel that follows the major categories of property and violent crimes that are consistently tracked by policymakers in Jamaica. Our research makes a significant contribution to the literature as one of the first studies to examine the impact of temperature and rainfall shocks on both property and violent crimes in a high-crime country that is located in Latin America and the Caribbean region. In particular, Jamaica is consistently ranked among the top 10 most violent countries in the world, with a homicide rate of about 58 victims per hundred thousand population ([Jamaica Constabulary Force 2022](#)).⁵ Consequently, our research provides valuable insights into the impact of weather shocks on criminal activity in developing countries that already contend with an elevated crime rate.

Several theories have outlined a direct mechanism that links the changes in weather conditions to changes in individuals behavior. First, the temperature–aggression or heat hypothesis suggests that when an individual is exposed to above-normal temperatures, they experience a greater degree of discomfort, triggering a psychological or physiological response that may manifest as increased aggression ([Harp and Karnauskas 2020](#); [Lynch, Stretesky, and Long 2020](#)). In other words, these external conditions may directly affect human judgment, causing heightened aggression and a loss of control ([Ranson 2014](#)).⁶ In our context, this theory suggests that higher temperatures may cause an increase in the number of violent crimes, though the predicted impact of temperature on property crime is less clear.⁷

Another theory that potentially explains the relationship between weather and crime is the routine activities theory. This theory suggests that the trends in crime are influenced by the patterns of people's daily activities ([Miró 2014](#)). According to the theory, for a crime to occur, it requires the presence of a motivated offender, a suitable target to victimize, and the absence of a capable guardian that could prevent or intervene to stop the crime ([Cohn and Rotton 2000](#)).⁸ Therefore, in the context of temperature, if an individual lacks access to home cooling technology, the routine activities theory predicts that warmer temperatures may increase the likelihood that they leave their homes unguarded and the risks associated with home break-ins ([Reichhoff 2017](#)). In contrast, since more rain potentially disrupts an individual's

5 This is important because Latin America and the Caribbean has one of the highest per capita crime rates in the world—about 20 victims per hundred thousand population. In comparison, the homicide rate in Asia and Africa is about 2.3 and 13 victims per hundred thousand population, respectively. To the best of our knowledge, the prior studies that have examined the weather–crime relationship in developing countries have focused on India and Mexico. The homicide rates in these countries are about 3 and 28, respectively. However, the homicide rate in Mexico is largely motivated by drug-related violence ([Baysan et al. 2019](#); [World Bank 2021](#)).

6 For example, studies have shown that professional baseball pitchers in the United States are more likely to purposefully hit batters with a pitched ball, and individuals are more likely to assertively “honk” their car horns during hotter days ([Lynch, Stretesky, and Long 2020](#); [Kenrick and MacFarlane 1986](#); [Larrick et al. 2011](#)).

7 The heat hypothesis suggests that increased temperatures can have a negative effect on individual's cognitive and emotional functioning, resulting in a higher likelihood of aggressive behavior. If the decision to engage in property crime is influenced by such responses, then it is possible that higher temperatures could also affect the incidence of these types of offences. However, there are alternative mechanisms through which higher temperature may affect crime. For instance, [Hsiang, Burke, and Miguel \(2013\)](#) argue that climatic changes may alter the supply of a resource and cause disagreement over its allocation, and climatic conditions may affect the relative preference of using violence or cooperation to achieve one's objectives.

8 In addition, similar to the routine activities theory, the social interaction theory of crime posits that the frequency of criminal acts is driven in large part by social interactions that occur in everyday life. When applied to weather, this hypothesis implies that weather conditions that encourage social interactions are more likely to increase crime rates ([Glaeser, Sacerdote, and Scheinkman 1996](#)). For example, mild weather conditions that encourage people to go shopping may increase the frequency of property crimes such as larceny ([Ranson 2014](#)).

daily routine and increases the likelihood that they stay at home, it may reduce the number of break-ins and larcenies that are observed.⁹

Consistent with the temperature–aggression hypothesis, we found that a one-standard-deviation increase in temperature (2°C) leads to a significant rise in violent crime, including murders (3.44 percent), shootings (7.53 percent), and incidents of aggravated assault (6 percent). On the other hand, while our results suggest that higher temperatures are weakly associated with a decrease in the number of rapes (3.41 percent), we find that it has no impact on property crime. This is likely because property crime is primarily determined by net economic gains rather than aggression. Additionally, we found that a one-standard-deviation increase in rainfall reduces the numbers of shootings (1.53 percent), break-ins (2.27 percent), and larcenies (3.85 percent), but it has no impact on any other categories of violent crime. These results are consistent with the routine activities theory of crime. We also find that our main results are consistent with alternative estimates obtained from the weighted least squares approach, Poisson and negative binomial models, and various spatial autoregressive models.

2. Background and Data

2.1. Weather Events and Crime in Jamaica

Jamaica is a middle-income developing country located in the Caribbean and is among the most vulnerable to the negative effects of climate change (Mertz et al. 2009). The country's susceptibility is impacted by constraints such as its high exposure to a range of natural hazards, the population's dependence on agricultural activities, and the concentration of settlements and infrastructure along low-lying coastal strips (Baptiste and Rhiney 2016; Reyer et al. 2017). Currently, about 90 percent of the country's GDP is within the coastal zone, resulting in its key industries and over half of the population being vulnerable to hurricanes, tropical storms, sea-level rise, and land loss (USAID Climate Change Support 2017). In addition, rising temperatures and heavy rainfall have increased the likelihood of low productivity and slow economic growth.

The increase in both air and sea-surface temperatures in the Caribbean has been linked to increased tropical storms and hurricane activities in the region, which will have an impact on freshwater resources, tourism, human health, and social security (World Bank 2023; Baptiste and Rhiney 2016). Between 2001 and 2012, Jamaica experienced 11 storm events (including 5 major hurricanes) and several flood and drought events, and between 2014 and 2015, the country experienced one of the worst seasons of drought in recent history, which had a devastating economic impact on rural livelihoods (Weller 2019).

These weather impacts are set to increase as USAID estimates that by 2050, average annual temperature will likely increase by 1.0–1.4°C and average annual rainfall will decrease by 4.8–7.2 percent. The number of “hot” days and nights and extreme rainfall days are also projected to increase by 52–59 percent and 3.1–14 percent, respectively (USAID Climate Change Support 2017). The presence of these weather-related stressors potentially increases the long-term impact of the effects of climate change and severe weather events (Agnew 2012; World Bank 2023).

On the other hand, Jamaica has one of the highest per capita crime rates in the world. The country recorded an average of 1,293 murders over the last 12 years, with 2017 being the highest in the country's history—recording 1,647 murders or about 58 murders per hundred thousand population (Jamaica Constabulary Force 2022). There are several social problems that have been found to correlate with the level of crime, including poor levels of youth development, high levels of unemployment, childbearing at a young age, poor or absent parenting, limited economic opportunities, and a lack of targeted psychological

9 Becker's theory of crime could also be utilized to explain the relationship between weather shocks and crime (Becker 1976). This theory suggests that individuals weigh the potential cost and benefits before committing a crime. As such, when weather conditions change, individuals may change their propensity towards certain crimes. For example, heavy rainfall may increase the risks associated with breaking and entering, as homeowners are more likely to stay at home.

support (Campbell et al. 2020; Harriott 2000). This study is one of the first to examine the interaction between weather shocks and criminal activities in this context.

2.2. Data

The crime data were obtained from the Jamaica Constabulary Force (JCF) Statistics Division. The JCF is the main agency that is responsible for collecting, verifying, and reporting on criminal activity in Jamaica. The agency consistently tracks the seven types/categories of crime that occur most frequently across the country. These major crime categories include murder, shooting, aggravated assault, rape, break-ins, larceny, and robbery. The main purpose of tracking these major crimes is to aid policymakers in understanding the state of crime in the country and to enable them to tailor social intervention and community development programs to address crime (Bourne, Brooks, and Quarrie 2023).

The collected records provide a census of all major crimes that are reported or founded by the police. This includes all incidents that were recorded through 119 emergency calls, walk-in reports to police stations, and discoveries made by police officers that were on patrol. To be included, each report must have been confirmed by the police. The data are collated at the victim level, with each incident potentially having multiple victims. For each victim, only the most severe offence is reported in the data.¹⁰ The information that is collected includes the time, date, and place each incident occurred, the type of criminal activity that was reported, any weapon that was utilized, and the police station that responded to the incident. Using this administrative data, we create a panel data set that tracks the seven major crimes each day within each police division. Following Blakeslee et al. (2021), we classify (a) murder, shooting, aggravated assault, and rape as violent crimes, and (b) break-ins, larceny, and robbery as property crimes in our analysis. The description of each of these major crimes is discussed in table 1. In our analysis, each of these crimes is expressed as a rate per hundred thousand population.¹¹

The weather data were collected from the Meteorological Service of Jamaica. These data were measured at the weather station by hour level.¹² The data include information on average air temperature, maximum and minimum temperatures, relative humidity, and rainfall. While all weather variables are recorded at the hourly level, we average these variables to the day level in our analysis. For instance, our rainfall variable measures the average hourly rainfall observed in a particular division on a given day. Lastly, we collected socio-economic data from the Planning Institute of Jamaica. These data include the population and the number of welfare recipients who are poor, students, elderly, and pregnant women in each location.¹³

- 10 Consistent with the laws of Jamaica, if someone is shot but survives, it is classified as a shooting incident in the data. However, if the person does not survive, it is recorded as a murder. Similarly, if a stabbing victim survives, it is considered an aggravated assault, but if the victim dies, it is also classified as murder. In addition, if a house burglar breaks in and stabs the homeowner who is still alive when the incident is reported, then this will be reported as an aggravated assault. It is also important to note that a single incident can involve multiple victims. For example, if an individual shoots four people and two of them die, this is counted as four victims—two cases of murder and two cases of shooting. Consequently, there is no double counting of victims who experienced multiple offences, as the most extreme offence supersedes any lesser offence(s). In addition, this data set does not include any drug-related offences and non-serious misdemeanors such as marijuana possession, noise complaints, or traffic offences.
- 11 The main advantage of using administrative crime data is that they are the most accurate measure of the crime that is available. In our case, the data we utilize are used to inform policymakers on the trends in major crime in the country. As such, the JCF takes great care to report these statistics accurately. While measurement error is still a possibility, we believe that such reporting errors will be non-systematic and uncorrelated with the temperature and rainfall level. If so, they should not bias our estimates.
- 12 There is at least one weather station located in each of the 14 parishes. As such, weather data are collected across the entire country. Given that the weather variables vary at the parish level, we re-estimate our main models at this level. These results are reported in table S1.3. The resulting estimates are consistent.
- 13 Figure S1.2 in the supplementary online appendix shows the variation in temperature and crime that we exploited in this paper. This figure shows how the main categories of crime vary with average temperatures across the country.

Table 1. Categories of Major Crimes

Offences	Definition
Murder	There is no simple statutory definition for murder in Jamaica; however, under common law on which Jamaica’s jurisprudence is based, murder is defined as the unlawful killing of any human being with malice aforethought.
Aggravated assault & shooting	While shooting and aggravated assault are both covered under the Wounding (S. 20) subsection of the Offences against the Person Act, the Jamaica Constabulary Force tracks and reports these two categories separately. The act states that “whosoever, shall unlawfully and maliciously, by any means whatsoever, wound, or cause any grievous bodily harm to any person, or shoot at any person, or by drawing a trigger, or in any other manner attempt to discharge any kind of loaded arms at any person, with the intent in any of the cases aforesaid, to maim, disfigure or disable any person, or with intent to resist or prevent the lawful apprehension or detainer of any person.” As such, any shooting offence that causes grievous bodily harm is tracked as <i>shooting</i> , while other forms of wounding with intent are grouped as <i>aggravated assault</i> .
Break-ins	The Jamaica Constabulary Force (JCF) categorizes burglary and other forms of breaking and entering offences under the main category break-in. The Offences against the Person Act is the legal basis for all such charges, which states that “every person who breaks and enters any dwelling-house, or any, building within the curtilage thereof and occupied therewith, or any schoolhouse, shop, warehouse, counting-house, office, store, garage, pavilion, factory, or workshop, or any building belonging to Her Majesty or to any Government department, or to any municipal or other public authority, and commits any felony therein.” It also defines burglary as follows: “Every person who in the night breaks and enters the dwelling-house of another with intent to commit any felony therein; or every person who in the night breaks out of the dwelling-house of another, having entered such dwelling-house with intent to commit any felony therein; or committed any felony in such dwelling-house, shall be guilty of a felony called burglary.” The related offences in the data set include burglary, sacrilege, shop, house, office, storehouse, and schoolhouse breaking.
Larceny	The Larceny Act (2015) defines a larceny as “a person who, without consent of the owner, fraudulently and without a claim of right made in good faith, takes and carries away anything capable of being stolen with intent, at the time of such taking, permanently to deprive the owner thereof.” This category includes larceny dwelling, larceny person, and larceny from motor vehicle.
Robbery	Under the Larceny Act (2015), an individual is guilty of the offence of robbery “being armed with any weapon or instrument, or being together with one other person or more, robs, or assaults with intent to rob any person. Every person who robs any person and, at the time of or immediately before immediately after such robbery, uses any personal violence to any person.”
Rape	The Sexual Offences Act states that “a man commits the offence of rape if he has sexual intercourse with a woman, without the woman’s consent; and knowing that the woman does not consent to sexual intercourse or recklessly not caring whether the woman consents or not.”

Source: Authors’ own compilation.
Note: Under the law of Jamaica, the definition and sentences for murder, shooting, and wounding with intent are outlined in the Offences against the Person Act. In addition, burglary and other forms of breaking and entering, larceny, and robbery are covered under the Larceny Act and rape is covered under the Sexual Offences Act of Jamaica. The Jamaica Constabulary Force consistently tracks each of the major crimes listed in the table. Each crime is classified based on the police report.

Together, the data form a balanced panel at the police-division-by-day level, over the period 2015–2021. The resulting sample size is 48,583 observations.¹⁴ We present the descriptive statistics of our main variables in [table 2](#). The table shows that, on average, there were about 0.45 crimes per hundred thousand population each day, with 0.23 of these incidents being gun related and 0.19 not involving any weapons. In addition, the table shows the mean and standard deviation for the various types of crimes and the socio-economic covariates that are included as controls in our models. Over the study period, each day,

14 There are 19 police divisions and 14 parishes. While there may be multiple police divisions in a given parish (i.e., Saint Catherine is divided into North and South Saint Catherine) a single police division cannot cut across multiple parishes. This gives a total police-division-by-day-level sample size of $N = 19 \times 7 \times 365 + 38 = 48,583$. Note that the 38 accounts for the 2 additional days that the 19 divisions observed during the leap years 2016 and 2020.

Table 2. Summary Statistics

	Mean	SD
Panel A: Victims breakdown		
All crimes	0.45	0.68
Gun involved	0.23	0.48
No weapon	0.19	0.43
8 a.m. to 5:59 p.m.	0.17	0.38
6 p.m. to 11:59 p.m.	0.17	0.41
12 a.m. to 7:59 a.m.	0.11	0.33
Panel B: Victims by type of crime		
Murder	0.10	0.29
Shooting	0.085	0.25
Agg. assault	0.04	0.19
Rape	0.04	0.18
Robbery	0.08	0.24
Break-ins	0.10	0.30
Larceny	0.013	0.10
Panel C: Covariates		
Air temperature (Celsius)	25.36	1.95
Rainfall (mm)	0.19	1.94
Humidity	81.62	9.91
Population	240,116.80	154,490.30
No. of welfare recipients: Students	14,787.82	8,020.51
No. of welfare recipients: Elderly	5,458.53	2,560
No. of welfare recipients: Poor	1,034.14	969.16
No. of welfare recipients: Women	3,016.97	1,630.59
Special programs participants	191.62	205.63
Observations	48,583	48,583

Source: Authors' own computation.

Note: The table presents the means and standard deviations for all dependent and explanatory variables utilized in our empirical models. Variables in panels A and B are reported as rates per hundred thousand population. While all weather variables are recorded at the hourly level, we average these variables to the day level in our analysis. For instance, our rainfall variable measures the average hourly rainfall observed in a particular division on a given day.

the average hourly temperature was 25.36°C, the average hourly rainfall was 0.19 mm, and the average hourly relative humidity was 81.62 percent.

3. Empirical Design

To estimate the effect of weather shocks on various measures of crime, we estimate the following econometric model:

$$\text{Crime}_{dt} = \alpha_0 + \text{Weather_Shocks}_{dt}\beta + X_{dy}\theta + \gamma_d + \gamma_{dow} + \gamma_{doy} + \gamma_m + \gamma_y + \epsilon_{dt},$$

where d indexes police division and t indexes time at the day-by-month-by-year level. The dependent variable, Crime_{dt} , is measured as the number of victims per hundred thousand population that is observed for a specific type of crime in division d at time t . We focus on seven major categories of crime: murder, shooting, aggravated assault, rape, break-ins, larceny, and robbery. The model controls for day-of-the-week (γ_{dow}), day-of-the-year (γ_{doy}), month-of-the-year (γ_m), year (γ_y), and police-division (γ_d) fixed effects. The model also controls for several socio-economic covariates, X_{dy} , such as popula-

tion density and the number of welfare recipients who are students, elderly, poor, and women in each location.¹⁵

Our explanatory variables of interest, $\text{Weather_Shocks}_{dt}$, represent the standardized variables of relative humidity, air temperature, and rainfall, each with a mean of 0 and standard deviation of 1. One-standard-deviation changes in temperature and rainfall correspond to about 2°C and 2 mm, respectively. The weather shocks capture the daily deviations from the long-term average weather condition in each geographic location. The weather shock variables capture day-to-day deviations from the long-term weather conditions in each geographic location.¹⁶ As such, conditional on the included covariates, our parameter of interest (β) has a causal interpretation if these shocks are independent of any omitted factors affecting violent and property crimes at the day level. Given the random variability of temperature and rainfall at the day level, this condition is likely satisfied. Nonetheless, we conduct several additional checks to demonstrate the robustness of our main findings.¹⁷

Lastly, to allow the error terms to be correlated within each police division on each calendar day, we cluster the standard errors by police division and day of the year. However, given that there are only 19 police divisions, a small cluster problem arises which may bias the cluster-robust standard errors and cause an over-rejection of the null hypotheses (Cameron, Gelbach, and Miller 2008). To address this concern, we compute wild bootstrap p -values for the two-way clustered standard errors. Both the standard errors and wild bootstrap p -values are reported beneath each estimate in our baseline results.

4. Empirical Results

In this section we present our main findings on the impact of weather shocks on various categories of crime. For the main results, the standard errors are two-way clustered by division and day of the year. The two-way clustered standard errors and wild bootstrap p -values are presented below each estimate in round and square brackets, respectively. All model specifications include police-division, day-of-the-week, day-of-the-year, month, and year fixed effects. We also control for humidity, population density, and the number of welfare recipients in each location who are students, elderly, poor, and women.

Overall, we find consistent evidence that a one-standard-deviation (1SD) increase in temperature increases violent crimes such as murders, shootings, and aggravated assaults. However, our results indicate that temperature changes have no meaningful impact on rape or property crimes such as break-ins, larceny, and robbery. On the other hand, we found that while a 1SD increase in rainfall reduces the number of shootings, break-ins, and larcenies, it has a minimal impact on other categories of violent and property crimes. We also show that these findings are consistent with the estimates from several sensitivity and robustness checks.

- 15 We can include both day-of-the-year and month-of-the-year fixed effects simultaneously in our model because the variation between leap and non-leap years ensures that day of the year is not perfectly correlated with month of the year. For instance, during a leap year, March 1st occurs on the 61st day of the year, while this date occurs on the 60th day of the year during a non-leap year.
- 16 These variables are measured as follows: $\text{Weather_Shocks}_{dt} = (W_{dt} - \bar{W}_d) / W_d^\sigma$. For instance, the temperature shock variable is calculated using the temperature observed in district d at time t minus the historical average day-level temperature observed in district d , divided by the standard deviation of the day-level temperature in district d . This is consistent with the way weather shocks are measured in the literature (Dimitrova and Muttarak 2020; Le and Nguyen 2021; Freudenreich, Aladysheva, and Brück 2022).
- 17 As expected, the various weather variables of interest (air temperature, rainfall, and humidity) are weakly correlated. As such, including or omitting one of these variables does not change the reported estimates. Given that humidity has no independent effect across our various specifications, we only report the results for rainfall and air temperature in our tables below. However, in table 6, we report the effect of heat index (a measure of feel-like air temperature) that combines both air temperature and humidity. This does not change our main findings.

Table 3. Impact of Temperature on Overall Criminal Activities

	No. of victims	Violent crime	Property crime	Gun	No weapons	8 a.m.–5:59 p.m.	6 p.m.–11:59 p.m.	12 a.m.–7:59 a.m.
Panel A:								
Temp shock	0.011** (0.007) [0.048]	0.011*** (0.004) [0.008]	0.0003 (0.006) [0.956]	0.011** (0.005) [0.0024]	0.001 (0.004) [0.742]	0.005* (0.003) [0.0063]	0.007*** (0.003) [0.009]	–0.001 (0.003) [0.796]
Panel B:								
Rainfall	–0.0025 (0.0035) [0.353]	–0.0012 (0.0029) [0.639]	–0.0022 (0.0022) [0.186]	–0.0025 (0.0020) [0.141]	0.0009 (0.0020) [0.582]	0.0003 (0.0022) [0.848]	0.0000 (0.0022) [0.974]	–0.0028* (0.0023) [0.069]
Temp est. Outcome mean	2.44	3.67	0.13	4.78	0.53	2.94	4.11	–0.91
Rainfall est. Outcome mean	–0.56	–0.40	–0.96	–1.09	0.47	0.18	0.00	–2.55
Outcome mean	0.45	0.30	0.23	0.23	0.19	0.17	0.17	0.11
Std. deviation	(0.68)	(0.56)	(0.46)	(0.48)	(0.43)	(0.38)	(0.41)	(0.33)
Observations	48,583	48,583	48,583	48,583	48,583	48,583	48,583	48,583

Source: Authors' analysis based on data collected from the Statistics and Information Management Unit (SIMU) of the Jamaica Constabulary Force and the Climate Branch of the Meteorological Service Division.

Note: All specifications include police-division, day-of-the-week, day-of-the-year, month, and year fixed effects. We also control for humidity, population density, and the number of welfare recipients in each location who are students, elderly, poor, and women. Both air temperature and rainfall are standardized to have a mean of 0 and variance of 1. Each outcome variable is measured as the number of reported incidents per hundred thousand population in the police division. The mean and standard deviation for each outcome are shown in the bottom panel. The standard errors are two-way clustered by police division and day of the year. The two-way clustered standard errors and wild bootstrap *p*-values are presented below each estimate in round and square brackets, respectively. **p* < 0.1, ***p* < 0.05, ****p* < 0.01.

4.1. Impact of Weather Shocks on Overall Criminal Activities

In table 3, we report our results on various measures of aggregated crime, including the total number of (a) reported incidents, (b) violent crimes, (c) property crimes, (d) gun-related incidents, (e) non-weapon related incidents, and the total number of incidents reported during (f) daytime hours (8 a.m. to 5:59 p.m.), (g) evening hours (6 p.m. to 11:59 p.m.), and (h) late night or early morning hours (12 midnight to 7:59 a.m.). All outcome variables are measured as rates per hundred thousand population in the police division.

Panel A shows that a 1SD increase in the average daily air temperature (about 2°C) increases the number of victims by 0.011 per hundred thousand population or 2.44 percent of the daily average number of incidents over the period. Furthermore, we find that this uptick in criminal activities is driven by an increase in violent crime, which increased by 0.011 victims per hundred thousand population (3.67 percent), or gun-related offences, which also increased by 0.011 victims per hundred thousand population (4.78 percent). In addition, the results indicate that temperature increases have the largest impact on incidents occurring during daytime hours (2.94 percent) and evening hours (4.11 percent), but they have no impact on incidents reported during late night hours. However, we found that temperature changes has no impact on property crime or the number of reported crimes that did not involve a weapon.

On the other hand, the estimates in panel B show that a 1SD increase in the average hourly rainfall each day (1.95 mm per hour) has no significant impact on the total number of reported incidents or the overall number of violent and property crimes. However, the results suggest that higher-than-normal rainfall reduces the number of crimes that were reported during late night hours by 0.0028 (2.55 percent).

4.2. Weather Shocks and Violent Crimes

In table 4, we present our estimates on the impact of temperature and rainfall on various types of violent crimes, including (a) murders, (b) shootings, (c) aggravated assaults, and (d) rapes. The results in panel A show that a 1SD increase in the average daily air temperature increases the number of murder victims by 0.0033 per hundred thousand population (3.44 percent), shooting victims by 0.0064 per hundred thousand population (7.53 percent), and victims of aggravated assault by 0.0024 per hundred thousand population (6 percent) each day. In addition, the results show that the number of rape cases decreased by

Table 4. Impact of Weather Shocks on Violent Crimes

	Murder	Shooting	Agg. assault	Rape
Panel A: Temp shock	0.0033* (0.0025) [0.095]	0.0064** (0.0033) [0.022]	0.0024* (0.0020) [0.089]	−0.0014 (0.0014) [0.259]
Panel B: Rainfall	−0.0002 (0.0022) [0.929]	−0.0013* (0.0009) [0.073]	0.0005 (0.0012) [0.590]	−0.0002 (0.0007) [0.963]
Temp est. Outcome mean (percent)	3.44	7.53	6.00	−3.41
Rainfall est. Outcome mean (percent)	−0.21	−1.53	1.25	−0.49
Mean	0.096	0.085	0.040	0.041
Std. deviation	(0.29)	(0.25)	(0.19)	(0.18)
Observations	48,583	48,583	48,583	48,583

Source: Authors’ analysis based on data collected from the Statistics and Information Management Unit (SIMU) of the Jamaica Constabulary Force and the Climate Branch of the Meteorological Service Division.

Note: All specifications include police-division, day-of-the-week, day-of-the-year, month, and year fixed effects. We also control for humidity, population density, and the number of welfare recipients in each location who are students, elderly, poor, and women. Both air temperature and rainfall are standardized to have a mean of 0 and variance of 1. Each outcome variable is measured as the number of reported incidents per hundred thousand population in the police division. The mean and standard deviation for each outcome are shown in the bottom panel. The standard errors are two-way clustered by police division and day of the year. The two-way clustered standard errors and wild bootstrap *p*-values are presented below each estimate in round and square brackets, respectively. **p* < 0.1, ***p* < 0.05, ****p* < 0.01.

0.0014 per hundred thousand population (3.41 percent), but this effect was not statistically significant at conventional levels. In contrast, panel B shows that a 1SD increase in average hourly rainfall each day reduced the number of shooting victims by 0.0013 per hundred thousand population or 1.53 percent. However, our results indicate that rainfall had no significant impact on the number of reported incidents of murder, aggravated assault, or rape over the period.¹⁸

Our findings are consistent with the temperature–aggression hypothesis, which states that as temperature rises, individuals have a greater propensity to display more aggression. In this respect, our results show that exposure to a higher temperature likely leads to more aggressive behavior, which is a precursor for violent crimes such as murders, shootings, and aggravated assaults. Similarly, the results for rainfall are consistent with the routine activity theory, since rainfall prevents large outdoor social gatherings which are hot spots for shootings in Jamaica. However, it seems that the interruption of individuals’ routines due to rainfall does not have a discernible impact on other categories of violent crime.

4.3. Weather Shocks and Property Crimes

Table 5 reports the estimated effects of temperature and rainfall on various property crimes, such as break-ins, larceny, and robbery.¹⁹ The estimates show that as the average daily air temperature increases, it has no significant effect on the number of break-ins, larcenies, and robberies that were reported to the police. These null results are also consistent with the heat–aggression hypothesis, because the primary incentive for committing a property crime is monetary gain, and aggression, if utilized, is mainly an instrument to

18 In tables S1.1 and S1.2 of the supplementary online appendix, we demonstrate that removing the socio-economic controls does not alter the main findings, suggesting that the general weather–crime relationship is not mediated by these variables. In addition, we present key results on the heterogeneous effects and the joint effects of temperature and rainfall on violent and property crimes in supplementary online appendices S2 and S3, respectively. Both appendices are self-contained, featuring the main tables and corresponding discussions.

19 While robbery could be considered a violent crime because of the threat or use of force during robberies, we classify it as a property crime to be consistent with the prior literature. This classification is inconsequential to the main arguments we outline moving forward, as we find no effect of temperature or rainfall on this type of crime.

Table 5. Impact of Weather Shocks on Property Crimes

	Break-ins	Larceny	Robbery
Panel A: Temp shock	−0.0005 (0.0018) [0.868]	0.0002 (0.0010) [0.969]	0.0005 (0.0027) [0.776]
Panel B: Rainfall	−0.0022** (0.0018) [0.049]	−0.0005*** (0.0002) [0.003]	0.0005 (0.0014) [0.585]
Temp est. Outcome mean (percent)	−0.52	1.54	0.63
Rainfall est. Outcome mean (percent)	−2.27	−3.85	0.63
Mean	0.097	0.013	0.079
Std. deviation	(0.30)	(0.10)	(0.24)
Observations	48,583	48,583	48,583

Source: Authors' analysis based on data collected from the Statistics and Information Management Unit (SIMU) of the Jamaica Constabulary Force and the Climate Branch of the Meteorological Service Division.

Note: All specifications include police-division, day-of-the-week, day-of-the-year, month, and year fixed effects. We also control for humidity, population density, and the number of welfare recipients in each location who are students, elderly, poor, and women. Both air temperature and rainfall are standardized to have a mean of 0 and variance of 1. Each outcome variable is measured as the number of reported incidents per hundred thousand population in the police division. The mean and standard deviation for each outcome are shown in the bottom panel. The standard errors are two-way clustered by police division and day of the year. The two-way clustered standard errors and wild bootstrap *p*-values are presented below each estimate in round and square brackets, respectively. **p* < 0.1, ***p* < 0.05, ****p* < 0.01.

obtain the property of the victim (Cohn and Rotton 2000). As such, while higher temperature may cause an individual to be more aggressive if they decide to commit a property crime, it should not induce an individual into committing this type of crime.²⁰

Panel B indicates that a 1SD increase in daily average hourly rainfall reduces the number of break-ins by 2.27 percent and larcenies by 3.85 percent, but it has no discernible impact on the number of robberies. This result corroborates the findings of Ranson (2014) and Lab and Hirschel (1988), who similarly found that precipitation reduces both larceny and break-ins. The results also align with the routine activities theory, as inclement weather tends to keep individuals at home, which reduces the likelihood that they are exposed to both types of crimes.²¹

4.4. Robustness Checks

In this section we systematically examine the robustness of the main results using several sensitivity and specification checks. For example, we assess whether our results are sensitive to various alternative model specifications, including (a) using alternative measures of temperature such as the heat index (feel-like temperature), minimum air temperature, and maximum air temperatures, (b) examining non-linear effects by temperature levels, and (c) re-estimating our baseline model at the parish level.

In addition, in supplementary online appendix S1, we present supplemental analyses that provide additional support for the robustness of our main findings. Specifically, we report results from several alternative estimation approaches, such as the weighted least squares and count outcome models (Poisson and

20 In contrast, for violent crime, higher temperatures lead to affective aggression, where the primary purpose is to cause the injury of another person (Cohn and Rotton 2000).

21 Since we are testing multiple hypotheses simultaneously, we adjust the *p*-values from tables 3 and 4 using the Bonferroni, Holm, and Sidak methods. These results are presented in table S1.3. When we adjust the *p*-values for multiple hypothesis testing, we find that temperature no longer has a statistically significant impact on murder, but all other significant relationships survive at least at the 10 percent level.

Table 6. Alternative Measures of Temperature

	Heat index	Max air temp	Min air temp
No. of victims	0.0096* (0.0065) [0.065]	0.0097* (0.0074) [0.075]	0.012** (0.0076) [0.048]
Murder	0.0029 (0.0026) [0.221]	0.0031 (0.0024) [0.165]	0.0033 (0.0025) [0.133]
Shooting	0.0059** (0.0031) [0.021]	0.0063** (0.0034) [0.036]	0.0064** (0.0033) [0.019]
Agg. assault	0.0023* (0.0017) [0.074]	0.0024* (0.0018) [0.093]	0.0020 (0.0019) [0.161]
Rape	−0.0014** (0.0013) [0.049]	−0.0014* (0.0015) [0.063]	−0.0017** (0.0014) [0.021]
Break-ins	−0.001 (0.003) [0.808]	−0.0007 (0.0035) [0.808]	0.0005 (0.0040) [0.870]
Larceny	0.00004 (0.001) [0.972]	0.0002 (0.001) [0.840]	0.0002 (0.001) [0.818]
Robbery	0.0007 (0.0028) [0.728]	−0.00001 (0.0026) [0.996]	0.0011 (0.0012) [0.589]
Observations	48,583	48,583	48,583

Source: Authors’ analysis based on data collected from the Statistics and Information Management Unit (SIMU) of the Jamaica Constabulary Force and the Climate Branch of the Meteorological Service Division.
Note: All specifications include police-division, day-of-the-week, day-of-the-year, month, and year fixed effects. We also control for humidity, population density, and the number of welfare recipients in each location who are students, elderly, poor, and women. All outcome variables are standardized to have a mean of 0 and variance of 1. Each outcome variable is measured as the number of reported incidents per hundred thousand population in the police division. The standard errors are two-way clustered by police division and day of the year. The two-way clustered standard errors and wild bootstrap *p*-values are presented below each estimate in round and square brackets, respectively. **p* < 0.1, ***p* < 0.05, ****p* < 0.01.

negative binomial). Also, in supplementary online appendix S4, we show that similar results are obtained when models that account for spatial autocorrelation are utilized.

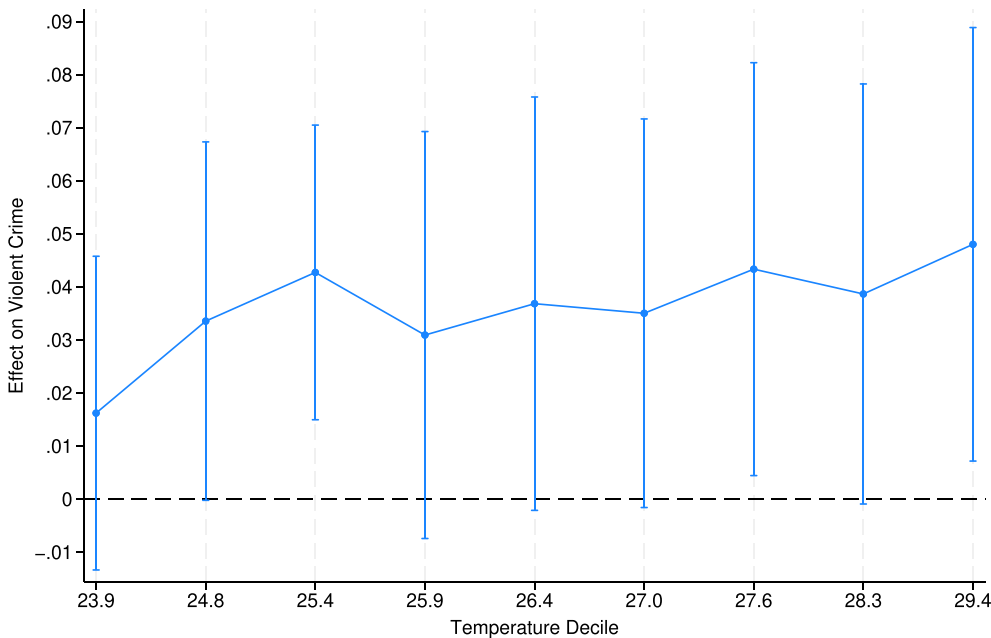
4.4.1. Alternative Specifications

In this subsection we examine the sensitivity of our results to alternative model specifications, such as (a) using alternative measures of temperature, (b) examining non-linear effects by temperature deciles, and (c) re-estimating our baseline model at the parish level.²²

In [table 6](#), we examine the sensitivity of our main result to the temperature measure we utilized. As such, while we used average recorded air temperature in our baseline models, we found that equivalent results would be obtained if heat index (a measure of feel-like temperature), the minimum recorded air

22 While Jamaica has 19 police divisions, the country is geographically divided into 14 parishes. Parishes are a slightly higher level of geographical aggregation.

Figure 1. The Effect of Temperature on the Number of Violent Crimes.



Source: Authors' own calculation.

Note: Figure showing the effect of temperature on the number of violent crime victims for each temperature decile. The point estimate is represented on the vertical axis, and the deciles for temperature are shown on the horizontal axis.

temperature, or the maximum recorded air temperature was instead utilized.²³ These estimates demonstrate that our baseline results are not sensitive to the way our temperature variable is measured.²⁴ Next, in [figs 1–3](#), we show the effect of temperature (by deciles) on the outcomes that were most affected in our baseline models: total violent crimes, murders, and shootings per hundred thousand population. The average temperature in each decile is shown on the horizontal axes. The results indicate that relative to temperatures in the lowest decile, temperatures in higher deciles are associated with higher rates of violent crime, a trend which is also observed for murder and shooting incidents. However, since the standard errors are quite large, the observed differences across deciles are not statistically significant. In particular, further analysis suggests that we fail to reject the null hypothesis that the estimated coefficients are statistically equal across temperature deciles.

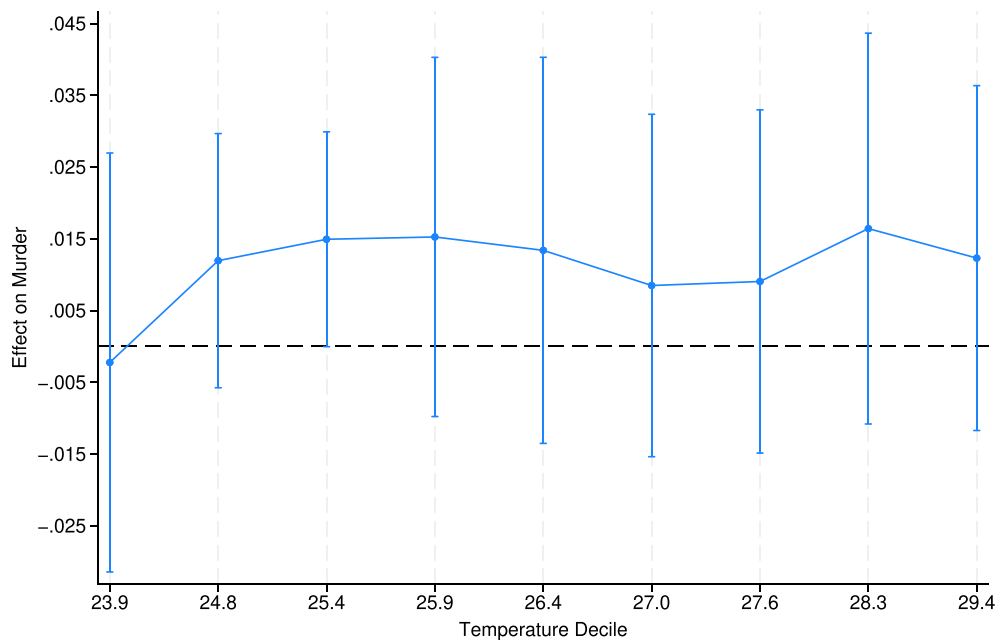
23 We utilize the formula for heat index from the National Oceanic and Atmospheric Administration. The formula for the index is as follows:

$$\begin{aligned} \text{Heat_Index} = & -42.379 + 2.04901523 * \text{Temp} + 10.14333127 * \text{RH} \\ & - 0.22475541 * \text{Temp} * \text{RH} - 0.00683783 * \text{Temp}^2 - 0.05481717 * \text{RH}^2 \\ & + 0.00122874 * \text{Temp}^2 * \text{RH} + 0.00085282 * \text{Temp} * \text{RH}^2 \\ & - 0.00000199 * \text{Temp}^2 * \text{RH}^2, \end{aligned}$$

where RH is relative humidity and Temp is measured in degrees Fahrenheit. We also utilized the proposed adjustments for extreme humidity and extreme temperatures.

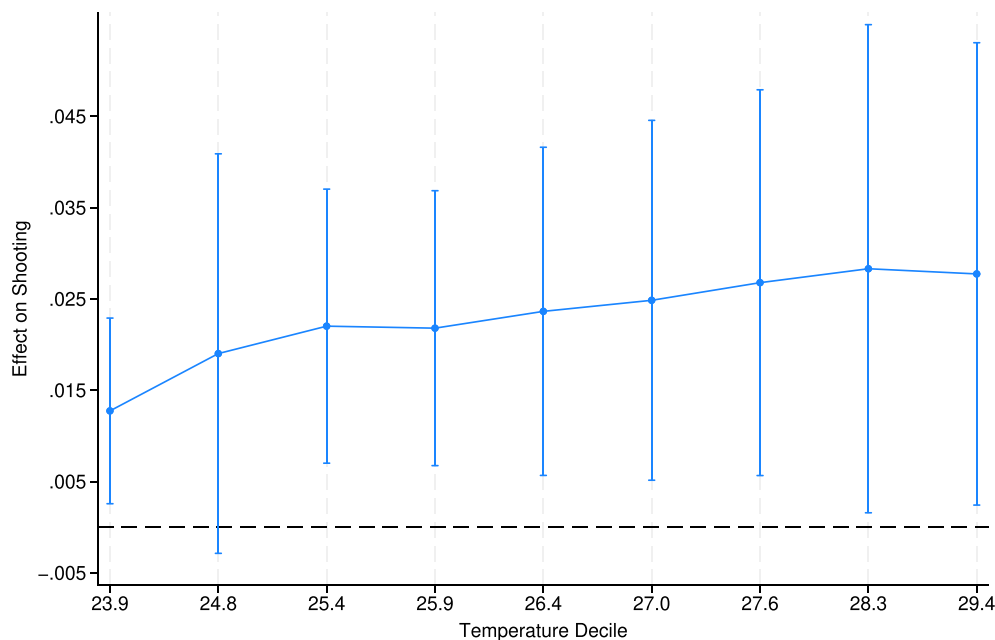
24 In [fig. S1.3](#) in the supplementary online appendix, we show that the impact of temperature varies with the level of humidity. For example, high levels of humidity can amplify the impact of temperature on violent crime. This explains the relationship between the heat index and violent crime.

Figure 2. The Effect of Temperature on the Number of Murder Victims.



Source: Authors' own calculation.
Note: Figure shows the effect of temperature on the number of murder victims for each temperature decile. The point estimates for murder is represented on the vertical axis, and the deciles for temperature are shown on the horizontal axis

Figure 3. The Effect of Temperature on the Number of Shooting Victims.



Source: Authors' own calculation.
Note: Figure showing the effect of temperature on the number of shooting victims for each temperature decile. The point estimate for shooting is represented on the vertical axis, and the deciles for temperature are shown on the horizontal axis.

Lastly, we examine the sensitivity of our main results to the level of geographic aggregation. Since police divisions are nested within parishes, we re-estimate our models at the parish level. The standard errors are two-way clustered by parish and day of the year for this model specification. These results are presented in [table S1.4](#) of the supplementary online appendix. Using this parish-by-day-level data set, we found larger estimates that are consistent with the higher level of aggregation. Nonetheless, these estimates are still generally consistent with our baseline findings.²⁵

4.4.2. Weighted Least Squares

Our main results in [tables 4–5](#) treat each observation equally, and for all estimates, we reported standard errors that were clustered at the division level. However, [Solon, Haider, and Wooldridge \(2015\)](#) note that when group-averaged data are utilized and the averages for different groups are based on widely varying within-group sample sizes, there are some important considerations that practitioners must take into account. First, they argue that one widely utilized justification for weighting is to correct for heteroskedasticity and obtain more precise estimates. However, depending on the structure of the error term, reweighting may produce less precise estimates in application. Therefore, the authors suggest that the appropriateness of reweighting is an empirical question. Next, [Solon, Haider, and Wooldridge \(2015\)](#) argue that both ordinary least squares (OLS) and weighted least squares (WLS) produce consistent estimates under exogenous sampling and correct specification of the regression model. As such, large differences between the OLS and WLS estimates can be used as a diagnostic tool for model misspecification or endogenous sampling. For these reasons, the authors conclude that it is good practice to present both the WLS and OLS estimates in practice.

Therefore, we use division-level population to estimate the WLS estimates and present these results in column 1 of [tables S1.5](#) and [S1.6](#) of the supplementary online appendix. The WLS estimates have a similar magnitude but a smaller standard error than the baseline results. Nonetheless, the statistical significance of both temperature and rainfall is consistent across the two empirical approaches.

4.4.3. Count Outcome Models

In our main results, for each specific type of crime, we measured the outcome variables as the number of reported victims per hundred thousand population in the police division. However, given that our outcomes of interest are all count variables that take on non-negative integer values, in this section we present alternative estimates of the impact of weather on crime using the Poisson and negative binomial models. We present these estimates in columns 2–3 of [tables S1.5](#) and [S1.6](#). The estimates of the Poisson model are quite similar to the baseline results. For instance, the results suggest that a 1SD increase in daily air temperature increases murder by 5.4 percent, shooting by 5.2 percent, and aggravated assault by 6.3 percent. These estimates are quantitatively consistent with the baseline estimates when expressed relative to the mean of the dependent variable. Similarly, consistent with the main results, the Poisson model suggests that rainfall reduced break-ins and larcenies by about 2.6 percent and 5.7 percent, respectively. However, these estimates are quite noisy and statistically insignificant, likely because of the lower frequency with which these types of crimes are observed. Since the estimates from the negative binomial model are al-

25 To further address concerns that our results may be explained by unobserved factors, in [tables S1.7](#) and [S1.8](#) we present the estimated impact of leads and lags of temperature and rainfall on the dependent variables that were most affected in the baseline model. In each model, we omit the first lead and lag to avoid issues of multicollinearity with the contemporaneous variables. As expected, we found that the leads and lags of temperature and rainfall have no impact on contemporaneous crimes. As such, we found that only contemporaneous weather shocks impacted crime, though some estimates are noisier due to the high degree of correlation in each variable over time.

most identical to the Poisson results, the findings from the count data models are very consistent with our baseline results.²⁶

4.5. The Way Forward

Our results suggest that both temperature and rainfall shocks have a meaningful impact on crime. Therefore, as the climate changes and weather patterns become more extreme, this will likely worsen the level of criminal activity in developing countries. For instance, our baseline results suggest that when the temperature increases by 2°C, this will increase the total number of victims by 2.44 percent, with murders increasing by 3.44 percent, shootings by 7.53 percent, and aggravated assaults by 6 percent each day. While a permanent increase of 2°C may seem high and unlikely, several climate change experts have projected that if emissions continue to rise, global temperatures are likely to increase by 1.1°C to 5.4°C by 2100 (Libertini 2014). In the case of Jamaica, USAID projects that the average annual temperature will likely increase by 1–1.4 degrees, and the number of hot days is projected to increase by over 50 percent by 2050. Therefore, as Jamaica and other developing countries experience warming over the next two decades, this may have a serious impact on the levels of violent crime.

Several mitigation strategies have the potential to disrupt the link between weather shocks and crime. Firstly, while awareness of the weather–crime relationship is slowly coming into sharper focus, the policy response at the national, sub-national, and municipal levels is lagging (Muggah 2022). As a result, it is important for policymakers to consider this relationship when designing and implementing crime-related policies.²⁷ Secondly, developing countries must map and assess the vulnerability and differentiated effects of weather-related shocks on specific population groups. For instance, Muggah (2022) argues that it is common for poorer minorities and underserved communities to face the most severe risks and egregious consequences from extreme weather events. In addition, developing countries are more vulnerable to severe weather shocks because they have a lower rate of climate technology adoption due to their financial constraints. Consequently, policymakers in developing countries should start considering the most cost-effective approach to adopting the necessary technology that can minimize the severe effects of future weather shocks.

5. Conclusion

In this study, consistent with the heat hypothesis, we found strong evidence that rising temperatures are linked to more violent crimes such as murder, shooting, and aggravated assault. However, while numerous studies have found that higher temperatures reduce property crime (Cohn and Rotton 2000; Ranson 2014), we found that temperature changes have no impact on the rates of break-ins, larcenies, and robberies. In the United States, where weather–crimes studies have predominantly been done, there are longer periods of cooler or winter-like temperatures and a general absence of consistent rainfall. In fact, Ranson (2014) found that the impact of temperature on property crime (larceny and break-ins) had a threshold effect, where temperature changes above 10°C had little effect on property crime rates. Consistent with these findings, our results indicate that in a country like Jamaica, which does not experience these low temperatures, there is no meaningful relationship between temperature and property crime. We also found that the level of rainfall has a minimal impact on violent crime (except shootings), but it reduces property crimes such as larcenies and break-ins. These results are consistent with the routine activities theory, which suggests that rainfall will likely disrupt some forms of social interactions and the crimes

26 While the Poisson and negative binomial models have the same mean structure, the latter has an additional parameter that accounts for overdispersion in the dependent variable (Wright and Dorilas 2022).

27 For instance, the evidence shows that increased property crime, intimate partner violence, and domestic abuse are linked to natural disasters (Muggah 2022). These dynamics must be taken into account in any disaster preparedness plans and targeted assistance initiatives.

that are often associated with them. Interestingly, we found no impact of weather conditions on violent sex crimes such as rape.

Consequently, even though Jamaica is consistently ranked among the most violent countries in the world, we have found consistent and robust evidence that weather shocks have a significant effect on the prevalence of violent and property crimes in the country. Since the global temperature is projected to increase by 1.1–5.4°C by 2100 and more extreme weather conditions are likely to become more frequent, our results suggest that there will be an increase in the level of criminal activity. Therefore, policymakers in Jamaica and other developing countries need to ponder the mitigation strategies that will be needed to combat higher temperatures and lower rainfall in the coming decades.

Data Availability Statement

The raw data can be obtained from the Statistics and Information Management Unit (SIMU) of the Jamaica Constabulary Force and the Climate Branch of the Meteorological Service Division, Ministry of Economic Growth and Job Creation (with permission).

References

- Acevedo, S., M. Mrkaic, N. Novta, E. Pugacheva, and P. Topalova. 2020. "The Effects of Weather Shocks on Economic Activity: What Are the Channels of Impact?" *Journal of Macroeconomics* 65: 103207.
- Agnew, R. 2012. "Dire Forecast: A Theoretical Model of the Impact of Climate Change on Crime." *Theoretical Criminology* 16(1): 21–42. <https://doi.org/10.1177/1362480611416843>.
- Althor, G., J. E. Watson, and R. A. Fuller. 2016. "Global Mismatch between Greenhouse Gas Emissions and the Burden of Climate Change." *Scientific Reports* 6(1): 20281.
- Baptiste, A. K., and K. Rhiney. 2016. "Climate Justice and the Caribbean: An Introduction." *Geoforum* 73: 17–21. <https://doi.org/10.1016/j.geoforum.2016.04.008>.
- Baysan, C., M. Burke, F. González, S. Hsiang, and E. Miguel. 2019. "Non-economic Factors in Violence: Evidence from Organized Crime, Suicides and Climate in Mexico." *Journal of Economic Behavior & Organization* 168: 434–52. <https://doi.org/10.1016/j.jebo.2019.10.021>.
- Becker, G. S. 1976. *The Economic Approach to Human Behavior*. University of Chicago Press.
- Binger, A. 2011. "Energy efficiency potential in Jamaica: Challenges, opportunities and strategies for implementation." Energy Efficiency Potential in Jamaica: Challenges, Opportunities and Strategies for Implementation. Retrieved from <https://repositorio.cepal.org/items/50220eac-4612-401b-9f69-a9dbe5094b58> on February 4, 2023.
- Blakeslee, D., R. Chaurey, R. Fishman, D. Malghan, and S. Malik. 2021. "In the Heat of the Moment: Economic and Non-economic Drivers of the Weather-Crime Relationship." *Journal of Economic Behavior & Organization* 192: 832–56. <https://doi.org/10.1016/j.jebo.2021.11.003>.
- Bourne, P. A., and D. M. Brooks. 2023. "A Quantitative Probe into Violent Crimes Committed in Jamaica from 2010 to 2022." *International Journal of Insights & Transformations in Law, Crime & Justice* 7(1): 16–36.
- Burke, M., S. M. Hsiang, and E. Miguel. 2015. "Global Non-linear Effect of Temperature on Economic Production." *Nature* 527(7577): 235–9. <https://doi.org/10.1038/nature15725>.
- Cameron, A. C., J. B. Gelbach, and D. L. Miller. 2008. "Bootstrap-Based Improvements for Inference with Clustered Errors." *Review of Economics and Statistics* 90(3): 414–27. <https://doi.org/10.1162/rest.90.3.414>.
- Campbell, H., P. Bourne, V. Peterkin, and M. Burke. 2020. "Politicking with Crimes in Jamaica: Specially Established Paramilitary Units in the Jamaica Constabulary Force, Zones of Special Operations, and States of Emergency." *The Corporate International [ISSN: 2581-6438 (online)]* 4(1): 1–88.
- Cohn, E. G., and J. Rotton. 2000. "Weather, Seasonal Trends and Property Crimes in Minneapolis, 1987–1988. A Moderator-Variable Time-Series Analysis of Routine Activities." *Journal of Environmental Psychology* 20(3): 257–72. <https://doi.org/10.1006/jevp.1999.0157>.
- Dimitrova, A., and R. Muttarak. 2020. "After the Floods: Differential Impacts of Rainfall Anomalies on Child Stunting in India." *Global Environmental Change* 64: 102130. <https://doi.org/10.1016/j.gloenvcha.2020.102130>.

- Freudenreich, H., A. Aladysheva, and T. Brück. 2022. "Weather Shocks across Seasons and Child Health: Evidence from a Panel Study in the Kyrgyz Republic." *World Development* 155: 105801.
- Garg, T., G. C. McCord, and A. Montfort. 2020. "Can Social Protection Reduce Environmental Damages?" <https://doi.org/10.2139/ssrn.3465356>.
- Glaeser, E. L., B. Sacerdote, and J. A. Scheinkman. 1996. "Crime and Social Interactions." *Quarterly Journal of Economics* 111(2): 507–48. <https://doi.org/10.2307/2946686>.
- Harp, R. D., and K. B. Karnauskas. 2020. "Global Warming to Increase Violent Crime in the United States." *Environmental Research Letters* 15(3): 034039. <https://doi.org/10.1088/1748-9326/ab6b37>.
- Harriott, A. 2000. *Police and Crime Control in Jamaica: Problems of Reforming Ex-colonial Constabularies*. University of West Indies Press.
- Hsiang, S. M., M. Burke, and E. Miguel. 2013. "Quantifying the Influence of Climate on Human Conflict." *Science* 341(6151): 1235367. <https://doi.org/10.1126/science.1235367>.
- Jamaica Constabulary Force. 2022. "JCF Statistics." Retrieved from <https://jcf.gov.jm/stats/> on February 2, 2023.
- Japan Refrigeration and Air Conditioning Industry Association. 2019. "World Air Conditioner Demand by Region." Retrieved from https://www.jraia.or.jp/english/World_AC_Demand.pdf on February 4, 2023.
- Kenrick, D. T., and S. W. MacFarlane. 1986. "Ambient Temperature and Horn Honking: A Field Study of the Heat/Aggression Relationship." *Environment and Behavior* 18(2): 179–91. <https://doi.org/10.1177/0013916586182002>.
- Lab, S. P., and J. D. Hirschel. 1988. "Climatological Conditions and Crime: The Forecast Is...?" *Justice Quarterly* 5(2): 281–99. <https://doi.org/10.1080/07418828800089731>.
- Larrick, R. P., T. A. Timmerman, A. M. Carton, and J. Abrevaya. 2011. "Temper, Temperature, and Temptation: Heat-Related Retaliation in Baseball." *Psychological Science* 22(4): 423–8. <https://doi.org/10.1177/0956797611399292>.
- Le, K., and M. Nguyen. 2021. "In-Utero Exposure to Rainfall Variability and Early Childhood Health." *World Development* 144: 105485. <https://doi.org/10.1016/j.worlddev.2021.105485>.
- Libertini, G. 2014. *Project for Powerful Wind Power Plants with Vertical Axis of Rotation: Verteolo Project*. Copernican editions.
- Lynch, M. J., P. B. Stretesky, and M. A. Long. 2020. "Climate Change, Temperature, and Homicide: A Tale of Two Cities, 1895–2015." *Weather, Climate, and Society* 12(1): 171–81.
- Mertz, O., K. Halsnæs, J. E. Olesen, and K. Rasmussen. 2009. "Adaptation to Climate Change in Developing Countries." *Environmental Management* 43: 743–752(2009). <https://doi.org/10.1007/s00267-008-9259-3>.
- Miró, F. 2014. "Routine Activity Theory." *The Encyclopedia of Theoretical Criminology*: 1–7. <https://doi.org/10.1002/9781118517390.wbetc198>.
- Mitchell, T. D., T. R. Carter, P. D. Jones, M. Hulme, and M. New 2004. "A Comprehensive Set of High-Resolution Grids of Monthly Climate for Europe and the Globe: The Observed Record (1901–2000) and 16 Scenarios (2001–2100)." *Tyndall Centre for Climate Change Research Working Paper* 55(0): 25.
- Muggah, R. 2021. "Climate Change and Crime in Cities." Retrieved from <https://igarape.org.br/en/climate-change-and-crime-in-cities/> on February 2, 2023.
- Parris, L.-A. J. 2023, "Map of Jamaica: Major Crimes and Temperature." Retrieved from Mona GeoInformatics (unpublished).
- Pavanello, F., E. De Cian, M. Davide, M. Mistry, T. Cruz, P. Bezerra, D. Jagu, S. Renner, R. Schaeffer, and A. F. Lucena. 2021. "Air-Conditioning and the Adaptation Cooling Deficit in Emerging Economies." *Nature Communications* 12(1): 6460.
- Racke, K., M. Skidmore, D. Hamilton, J. Unsworth, J. Miyamoto, and S. Cohen. 1997, "Pesticides Report 38. Pesticide Fate in Tropical Soils (Technical Report)." *Pure and Applied Chemistry - PURE APPL CHEM* 69: 1349–72. <https://doi.org/10.1351/pac199769061349>.
- Rahim, F. S., S. von Thadden-Kostopoulos, and C. Mungenast. 2023. "Islamic Republic of Mauritania: Climate Public Investment Management Assessment." *High-Level Summary Technical Assistance Reports* 2023(017).
- Ranson, M. 2014. "Crime, Weather, and Climate Change." *Journal of Environmental Economics and Management* 67(3): 274–302. <https://doi.org/10.1016/j.jeem.2013.11.008>.
- Reichhoff, M. 2017. *The Effect of Weather on Crime: An Investigation of Weather and Annual Crime Rates*. PhD thesis, University of Wisconsin–Whitewater.

- Reyer, C. P., S. Adams, T. Albrecht, F. Baarsch, A. Boit, N. Canales Trujillo, M. Carlsburg, D. Coumou, A. Eden, E. Fernandes *et al.*. 2017. "Climate Change Impacts in Latin America and the Caribbean and their Implications for Development." *Regional Environmental Change* 17: 1601–21(2017). <https://doi.org/10.1007/s10113-015-0854-6>.
- Ritchie, H., M. Roser, and P. Rosado. 2020. "CO and Greenhouse Gas Emissions." *Our World in Data*.no
- Ross, B., and B. McNary. 2023 . "Nearly 90 Percent of U.S. Households Used Air Conditioning in 2020." *U.S. Energy Information Administration*. Retrieved from <https://www.eia.gov/todayinenergy/detail.php?id=52558#>on February 2, 2023.
- Sachs, J. D. 2001. "Tropical Underdevelopment." <https://doi.org/10.3386/w8119>.
- Solon, G., S. J. Haider, and J. M. Wooldridge. 2015. "What Are We Weighting For?" *Journal of Human Resources* 50(2): 301–16. <https://doi.org/10.3368/jhr.50.2.301>.
- Statistical Institute of Jamaica. 2011. "Statistical Institute of Jamaica: Population 14 Years Old and Over Unemployed in the Week Preceding the Census by Activity Status by Parish by Sex." Retrieved from <https://statinja.gov.jm/Census/PopCensus/Population> on July 10, 2023.
- USAID Climate Change Support. 2017, "Climate Risk in Jamaica: Country Risk Profile (Task Order No. AID-OAA-TO-15-00030)." Retrieved from https://www.climatelinks.org/sites/default/files/asset/document/2017_USAID-CCIS_Climate-Risk-Profile-Jamaica.pdf on July 10, 2023.
- Weller, K. 2019, "Still Standing, Still Serving in the Midst of Extreme Weather." Retrieved from <https://www.usaid.gov/jamaica/news/still-standing-still-serving-midst-extreme-weather> on July 10, 2023.
- World Bank. 2021. "UN Office on Drugs and Crime's International Homicide Statistics." Retrieved from: <https://data.worldbank.org/indicator/VC.IHR.PSRC.P5> on July 10, 2023.
- World Bank. 2023. "Climate Change Knowledge Portal: Jamaica." Retrieved from <https://climateknowledgeportal.worldbank.org/country/jamaica/vulnerability> on July 10 2023.
- Wright, N. A., and E. Dorilas. 2022. "Do Cellphone Bans Save Lives? Evidence from Handheld Laws on Traffic Fatalities." *Journal of Health Economics* 85: 102659. <https://doi.org/10.1016/j.jhealeco.2022.102659>.